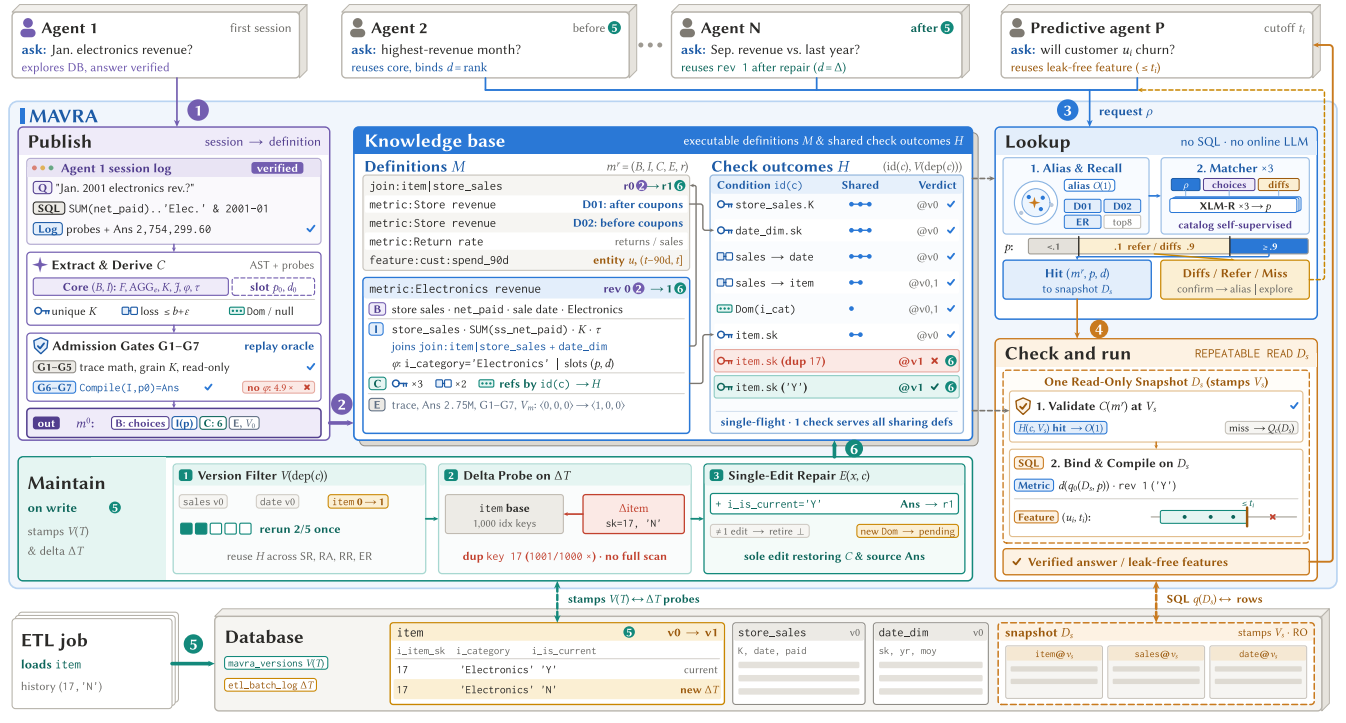

Agent 1
first session
ask: Jan. electronics revenue?
explores DB, answer verified
Agent 2
before 5
ask: highest-revenue month?
reuses core, binds d = rank
Agent N
after 5
ask: Sep. revenue vs. last year?
reuses rev 1 after repair (d = Δ)
Predictive agent P
cutoff t_i
ask: will customer u_i churn?
reuses leak-free feature (≤ t_i)
MAVRA
1
Publish
session → definition
Agent 1 session log
verified
Q "Jan. 2001 electronics rev.?"
SQL SUM(net_paid)..'Elec.' & 2001-01
Log probes + Ans 2,754,299.60
Extract & Derive C
AST + probes
Core (B, I): F, AGG_G, K, J, φ, τ
slot p_0, d_0
unique K
loss ≤ b+ε
Dom / null
Admission Gates G1–G7
replay oracle
G1–G5 trace math, grain K, read-only
G6–G7 Compile(I,p0)=Ans
no φ: 4.9 ×
out m^0: B: choices I(p) C: 6 E, V_0
2
Knowledge base
executable definitions M & shared check outcomes H
Definitions M
$m^r = (B, I, C, E, r)$
join:item|store_sales
r0 2 → r1 6
metric:Store revenue
D01: after coupons
metric:Store revenue
D02: before coupons
metric:Return rate
returns / sales
feature:cust:spend_90d
entity u_i (t−90d, t]
metric:Electronics revenue
rev 0 2 → 1 6
B store sales · net_paid · sale date · Electronics
I store_sales · SUM(ss_net_paid) · K · τ
joins join:item|store_sales + date_dim
φ: i_category='Electronics' | slots (p, d)
C ×3 ×2 refs by id(c) → H
E trace, Ans 2.75M, G1–G7, V_m: ⟨0, 0, 0⟩ → ⟨1, 0, 0⟩
Check outcomes H
(id(c), V(dep(c)))
Condition id(c) | Shared | Verdict
store_sales.K @v0
date_dim.sk @v0
sales → date @v0
sales → item @v0,1
Dom(i_cat) @v0,1
item.sk @v0
item.sk (dup 17) @v1 ×
item.sk ('Y') @v1 ✓
single-flight · 1 check serves all sharing defs
3
request ρ
Lookup
no SQL · no online LLM
1. Alias & Recall
alias O(1)
D01 D02
ER top8
2. Matcher ×3
ρ choices diffs
XLM-R ×3 → p
catalog self-supervised
p: ≤.1 .1 refer / diffs .9 ≥.9
Hit (m^r, p, d)
to snapshot D_s
Diffs / Refer / Miss
confirm → alias | explore
4
Check and run
REPEATABLE READ D_s
One Read-Only Snapshot D_s (stamps V_s)
1. Validate $C(m^r)$ at V_s
H(c, V_s) hit → O(1)
miss → $Q_c(D_s)$
SQL 2. Bind & Compile on D_s
Metric d($q_0(D_s, p)$) · rev 1 ('Y')
Feature (u_i, t_i):
≤ t_i
Verified answer / leak-free features
6
Maintain
on write 5
stamps V(T)
& delta ΔT
1 Version Filter V(dep(c))
sales v0
date v0
item 0 → 1
rerun 2/5 once
reuse H across SR, RA, RR, ER
2 Delta Probe on ΔT
item base
1,000 idx keys
Δitem
sk=17, 'N'
dup key 17 (1001/1000 ×) · no full scan
3 Single-Edit Repair E(x, c)
+ i_is_current='Y'
Ans → r1
≠ 1 edit → retire ⊥
new Dom → pending
sole edit restoring C & source Ans
stamps V(T) ↔ ΔT probes
SQL $q(D_s)$ ↔ rows
ETL job
loads item
history (17, 'N')
5
Database
mavra_versions V(T)
etl_batch_log ΔT
item
5
v0 → v1
i_item_sk i_category i_is_current
17 'Electronics' 'Y' current
17 'Electronics' 'N' new ΔT
store_sales v0
K, date, paid
date_dim v0
sk, yr, moy
snapshot D_s
stamps V_s · RO
item@v_s
sales@v_s
date@v_s